In this example, the model completes the full sequence of steps for $x = 51290.0$ and $y = 90718.0$. To assess the robustness of the Chain-of-Thought (CoT) process, we perform the following procedure for the Math Shepherd results:

1. Step-wise Completion: We cut off the generation after each step in the reasoning process. For instance, after computing $3x = 153770.0$, we stop the generation there and generate 10 completions using the Llama-3-8b model.
2. Multiple Completions: At each cut-off point, the Llama-3-8b model is tasked with completing the remaining steps of the chain of reasoning. For each step, 10 independent completions are generated.
3. Proportion of Correct Completions: For each cut-off point, we compute the proportion of correct completions. This proportion gives insight into how likely the model is to complete the remaining steps of reasoning correctly, starting from the intermediate point. For example, after cutting off the reasoning at $3x = 153770.0$, we evaluate how many of the 10 completions successfully compute $3x + 2y = 335306.0$.

In this way, Math-Shepherd quantifies the model's ability to continue reasoning correctly at each intermediate stage.

C.2.3 Additional results

Figures 8 - 10 present the sample-wise trajectories for 15 randomly chosen prompts leading to incorrect final answers for the different baselines. Here, once again, any significant drop in the plotted value at a given step could be seen as an indication of an incorrectly executed sub-task. Recall that in this setup majority of the errors occur at the final step which involves the addition of $3x + 2y$.

Figure 8 shows the sample-wise information-gain for our method after each step. We see that for most of the plotted trajectories, the sample-wise information-gain remains high until the final step, at which point it drops to values close to or below 0. This shows that our method correctly identifies that the failure predominantly occurs at step 3.

In contrast, Figure 9 shows that the mean probability of correctness for the ORM remains unchanged at each step. This could be explained by Figure 4 in the main text, which suggests that the ORM classifier can predict the correctness of the final output using only the values of x and y available in the prompt. Crucially, the classifier's confidence remains unchanged even as the model's intermediate reasoning steps are added to the input. This means that ORM is unable to distinguish between the model's performance on intermediate reasoning steps.

For Math-Shepherd results shown in Figure 10, most of the trajectories plotted remain constant at 0. In other words, when using Llama-3-8B as the completer model, we observe that for most of the prompts, no completion leads to the correct answer, regardless of which step we begin the completions from. This is likely because, for most of the examples considered in this plot, the (x, y) combination in the prompt has exactly one small value and the other is large (as shown in Figure 4). This also highlights why Math-Shepherd has a high false positive rate.

C.3 Controlled GSM8K Experiments

In order to understand if our proposed method also works on more textual data, we set out to perform an experiment on the popular GSM8K dataset which has more elaborate prompts compared to the previous experiments. To this end, we follow these steps:

- We first construct the dataset by using the GPT-4 API on the question. This will give us the basis for correct CoTs.
- Next we also again use GPT-4 to label each of the intermediate steps as either using "Addition", "Subtraction", "Division" or "Multiplication".
- With this data in hand, we can now construct our unidentifiable operations. In particular, we again use GPT-4 to rewrite all the CoTs which contain a multiplication into CoTs, where the multiplication is

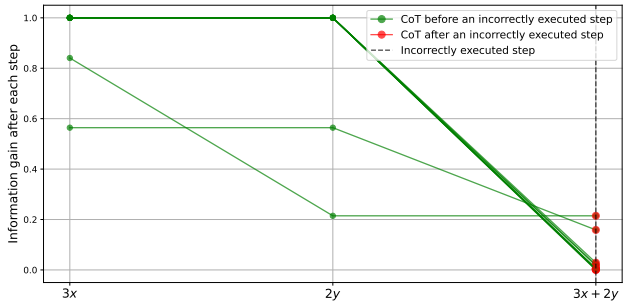

Figure 8 Arithmetic operations on Llama-3-8b: Sample-wise information-gain trajectories for 15 randomly chosen prompts with wrong final answers.

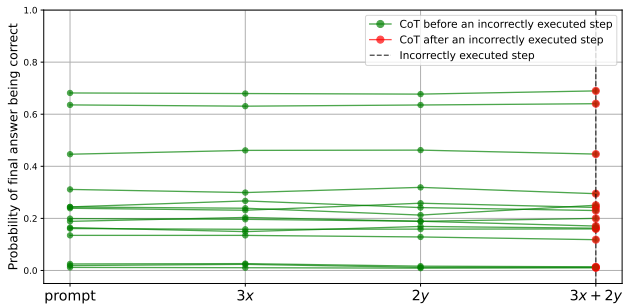

Figure 9 Arithmetic operations on Llama-3-8b: ORM's probability of correctness after each step for 15 randomly chosen prompts with wrong final answers.

performed wrong and subsequently also the final answer is wrong.

- Finally, we filter the final dataset, where we make sure that for every failed CoT, we have both "multiplication" and "subtraction" in the CoT, and for every correct CoT, we do not have these operations. This leads us to a setting where standard methods such as ORM would likely fail as there is no way to distinguish between multiplication and subtraction for the classifier.

C.3.1 Examples

Here below are some examples from the dataset, where we start with the prompt, followed by the CoT generations (with a wrong multiplication).

Prompt 1:

```
Julie is reading a 120-page book.  Yesterday, she was able to read 12 pages and today, she
read twice as many pages as yesterday.  If she wants to read half of the remaining pages tomorrow,
how many pages should she read?
```

Model response 1:

```
Calculate today's pages read by doubling yesterday's pages:  12 * 2 = 46.  Calculate the total
pages read so far by adding yesterday's and today's pages:  12 + 46 = 58.  Determine the remaining
pages by subtracting the pages read so far from the total:  120 - 58 = 62.  Calculate the number
of pages Julie should read tomorrow by taking half of the remaining pages:  62 / 2 = 31.
```

Prompt 2:

```
In a conference room, 40 chairs with a capacity of 2 people each were arranged in rows in preparation
for the board meeting of a company, whose number of members was the same as the chairs' capacity.
```

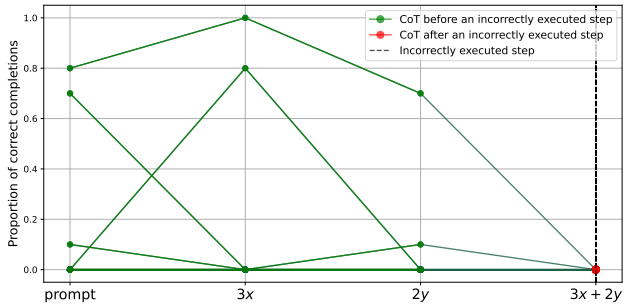

Figure 10 Arithmetic operations on Llama-3-8b: Math-Shepherd's proportion of correct completions from each step for 15 randomly chosen prompts with wrong final answers.

`If 2/5 of the chairs were not occupied, and the rest each had two people, calculate the number of board members who did attend the meeting.`

Model response 2:

`Calculate the total capacity of the conference room: 40 chairs * 2 people per chair = 134 people. Calculate the number of chairs not occupied: 40 chairs * 2/5 = 16 chairs. Calculate the number of chairs occupied: Total number of chairs - Chairs not occupied = 40 chairs - 16 chairs = 24 chairs. Calculate the number of people that attended (each occupied chair has 2 people): 24 chairs * 2 people per chair = 63 people.`

Prompt 3:

`Dan owns an ice cream shop and every sixth customer gets a free ice cream cone. Cones cost $2 each. If he sold $100 worth of cones, how many free ones did he give away?`

Model response 3:

`Calculate the number of cones sold that were not free by dividing the total sales by the cost per cone: 100 / 2 = 50. Calculate the total number of cones including free ones by assuming every sixth cone is free, so for every 5 sold, there is 1 free. Total cones = (5 sold + 1 free) * (cones sold / 5) = 6 * (50 / 5) = 6 * 10 = 72. Calculate the number of free cones given away: Total cones - Cones sold = 72 - 50 = 22.`

C.3.2 Training data for the supervisor model

For our supervisor model, we simply use a GPT-2 model that we SFT until convergence and use early stopping based on a held out validation dataset. The training data for this model is composed of generated CoT steps concatenated with the correct final output (as in other experiments).

For example, consider prompt 3 and its response above. For this prompt, the correct final response is 10. Using this prompt, we generate 4 training datapoints for the supervisor model by truncating the response at each step and concatenating the correct final answer using the separator token '`#|>`'.

1. `Dan owns an ice cream shop and every sixth customer gets a free ice cream cone. Cones cost $2 each. If he sold $100 worth of cones, how many free ones did he give away? #|> 10`
2. `Dan owns an ice cream shop and every sixth customer gets a free ice cream cone. Cones cost $2 each. If he sold $100 worth of cones, how many free ones did he give away? || Calculate the number of cones sold that were not free by dividing the total sales by the cost per cone: 100 / 2 = 50 #|> 10`
3. `Dan owns an ice cream shop and every sixth customer gets a free ice cream cone. Cones cost $2 each. If he sold $100 worth of cones, how many free ones did he give away? ||`